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Mercer

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(54) **LOUVERED FIN**

(71) Applicant: **Rheem Manufacturing Company**, Atlanta, GA (US)

(72) Inventor: **Kevin Mercer**, Danville, IN (US)

(73) Assignee: **Rheem Manufacturing Company**, Atlanta, GA (US)

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F28F 1/32 (2006.01)

(52) **U.S. Cl.**
CPC ***F28F 1/325*** (2013.01)

(58) **Field of Classification Search**
CPC F28F 1/325; F28F 1/128; F28F 2215/08; F28F 2215/10; F28F 1/32; F25B 39/02; F28D 1/05333
See application file for complete search history.

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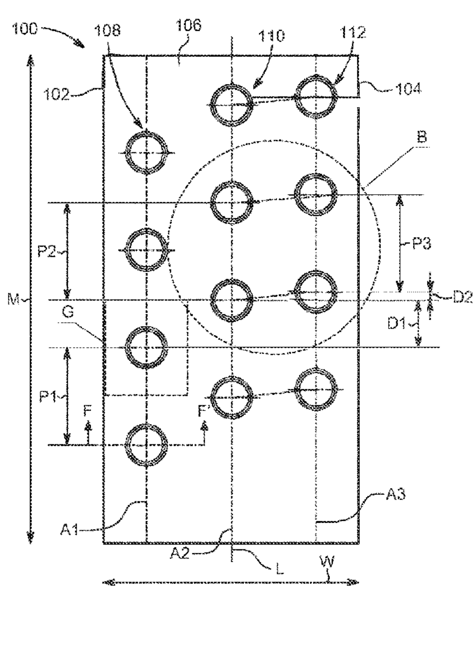
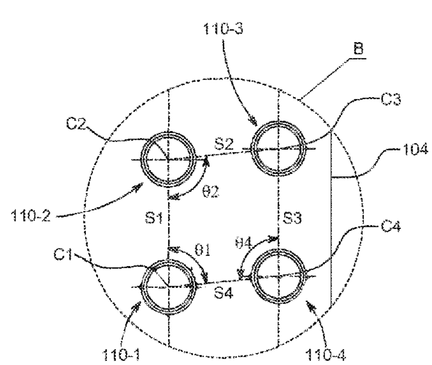
Primary Examiner — Len Tran
Assistant Examiner — Kamran Tavakoldavani
(74) *Attorney, Agent, or Firm* — Eversheds Sutherland (US) LLP

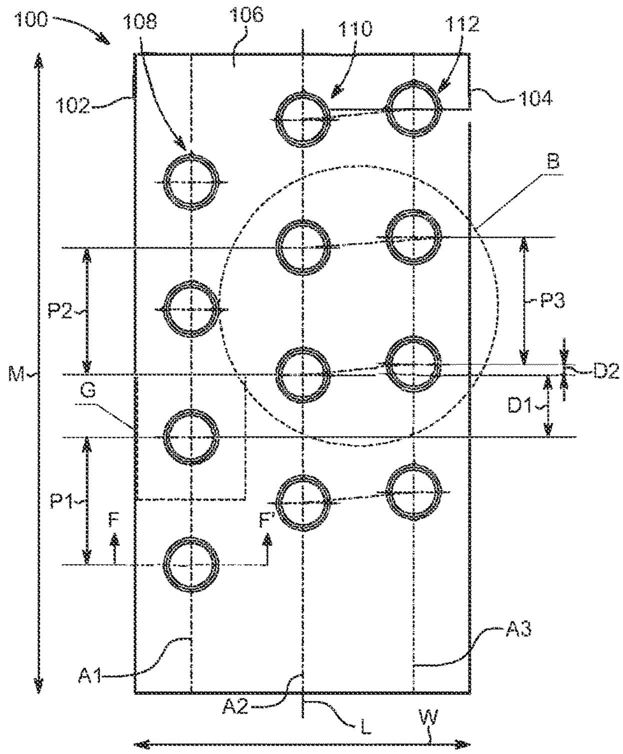
(57) **ABSTRACT**

The present disclosure provides a louvered fin including a leading edge, a trailing edge, and a surface extending between the leading edge and the trailing edge. The surface defines a first set of holes along a first axis, a second set of holes along a second axis and offset from the first set of holes, and a third set of holes along a third axis and offset from the second set of holes. Each of the first axis, the second axis, and the third axis extends substantially parallel to a longitudinal axis of the fin. A first offset distance between the second and first set of holes is greater than a second offset distance between the third and second set of holes. The second and the third set of holes define a substantially obtuse trapezoidal matrix.

23 Claims, 13 Drawing Sheets



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106
110
112
102
104
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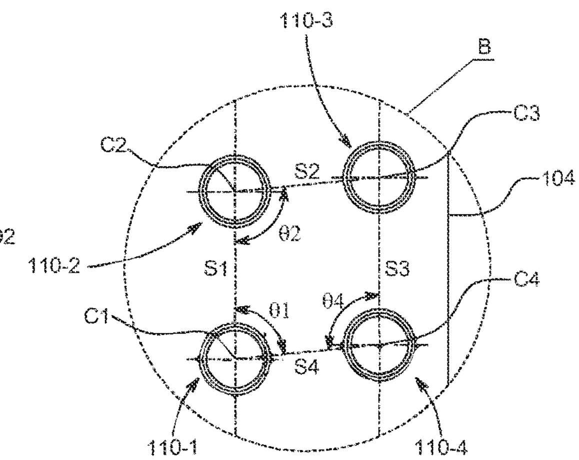
P3
D2
M
G
D1
P1
F
F'
A3
A1
A2
L
W

FIG. 1A


110-3
B
C3
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S2
104
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110-2
S1
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C4
θ1
θ4
C1
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110-1
110-4

FIG. 1B

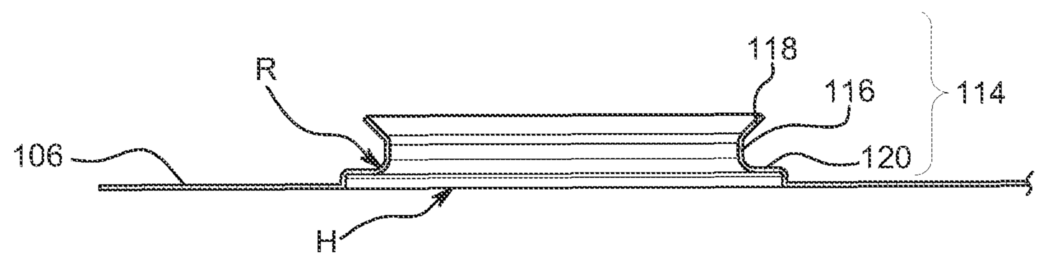

FIG. 1C

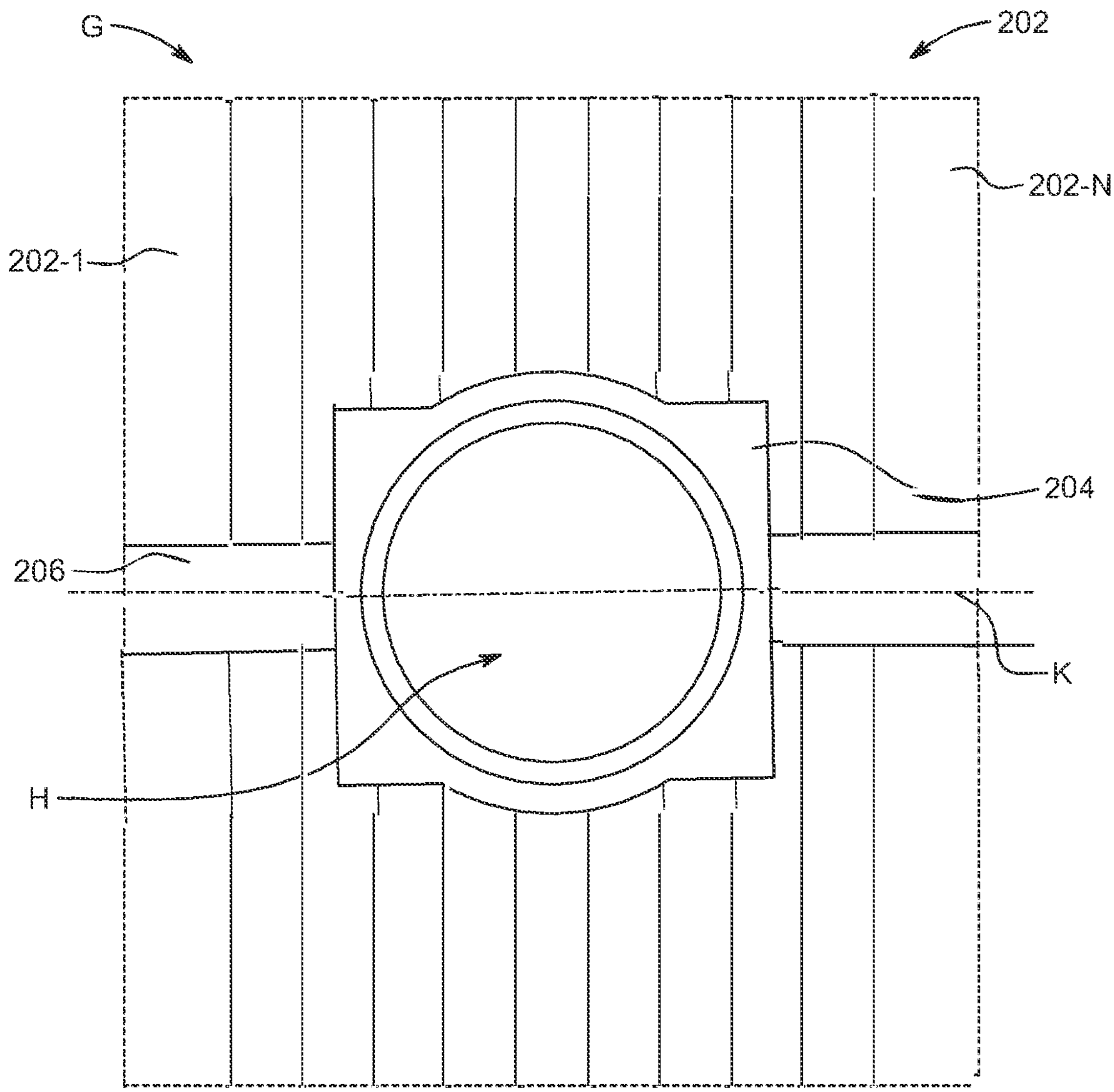

FIG. 2

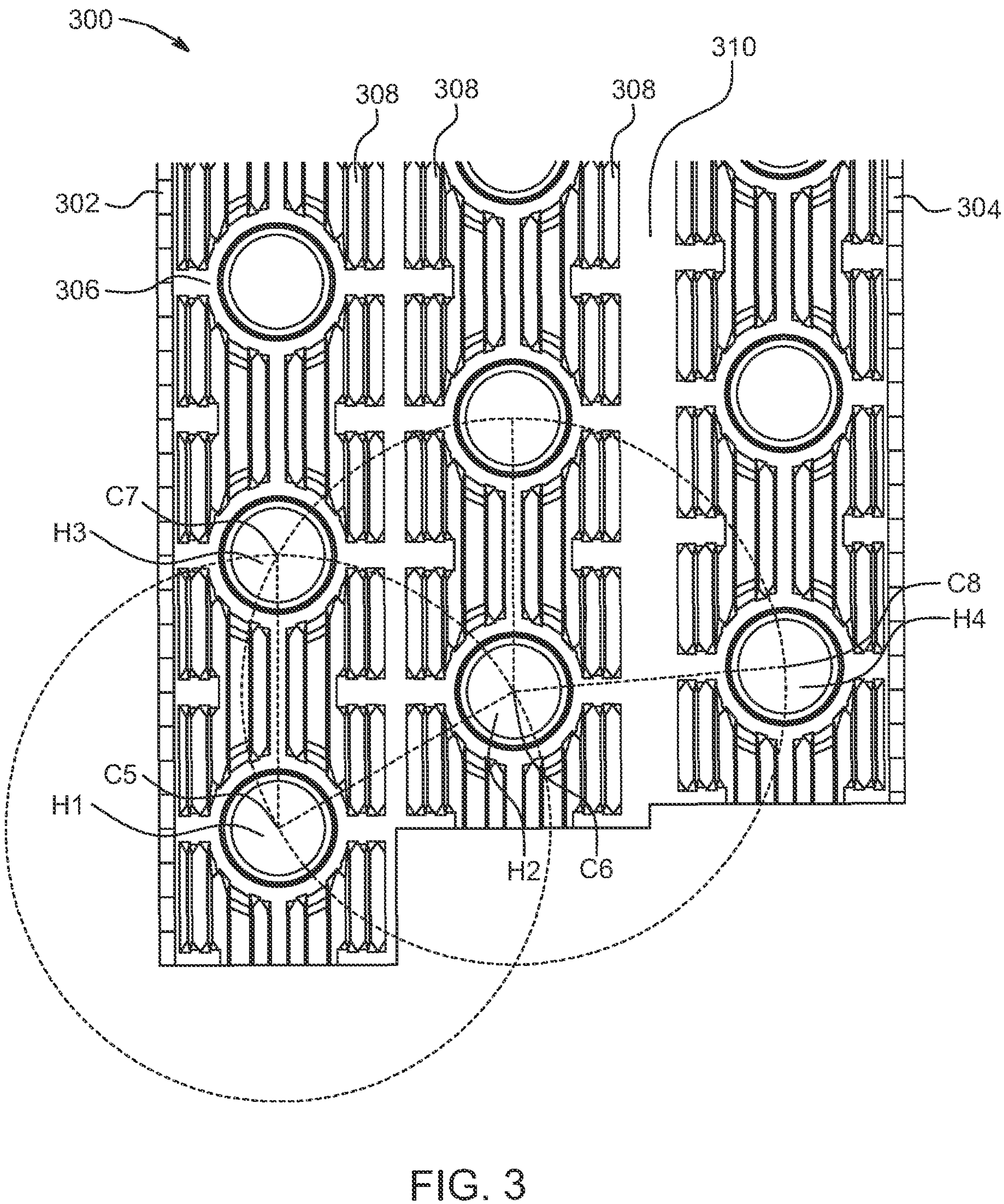

FIG. 3

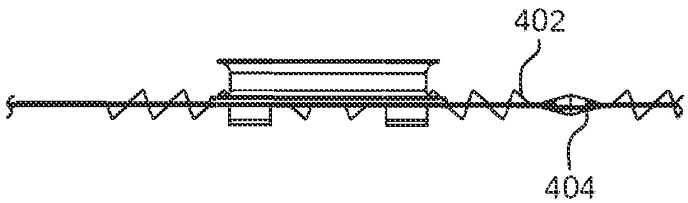

FIG. 4

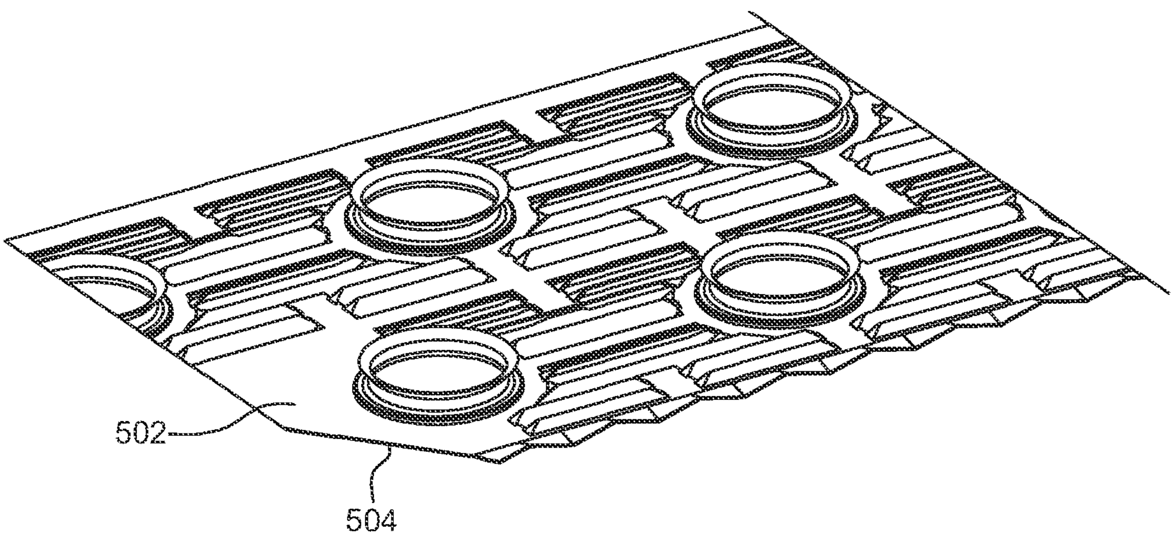

FIG. 5

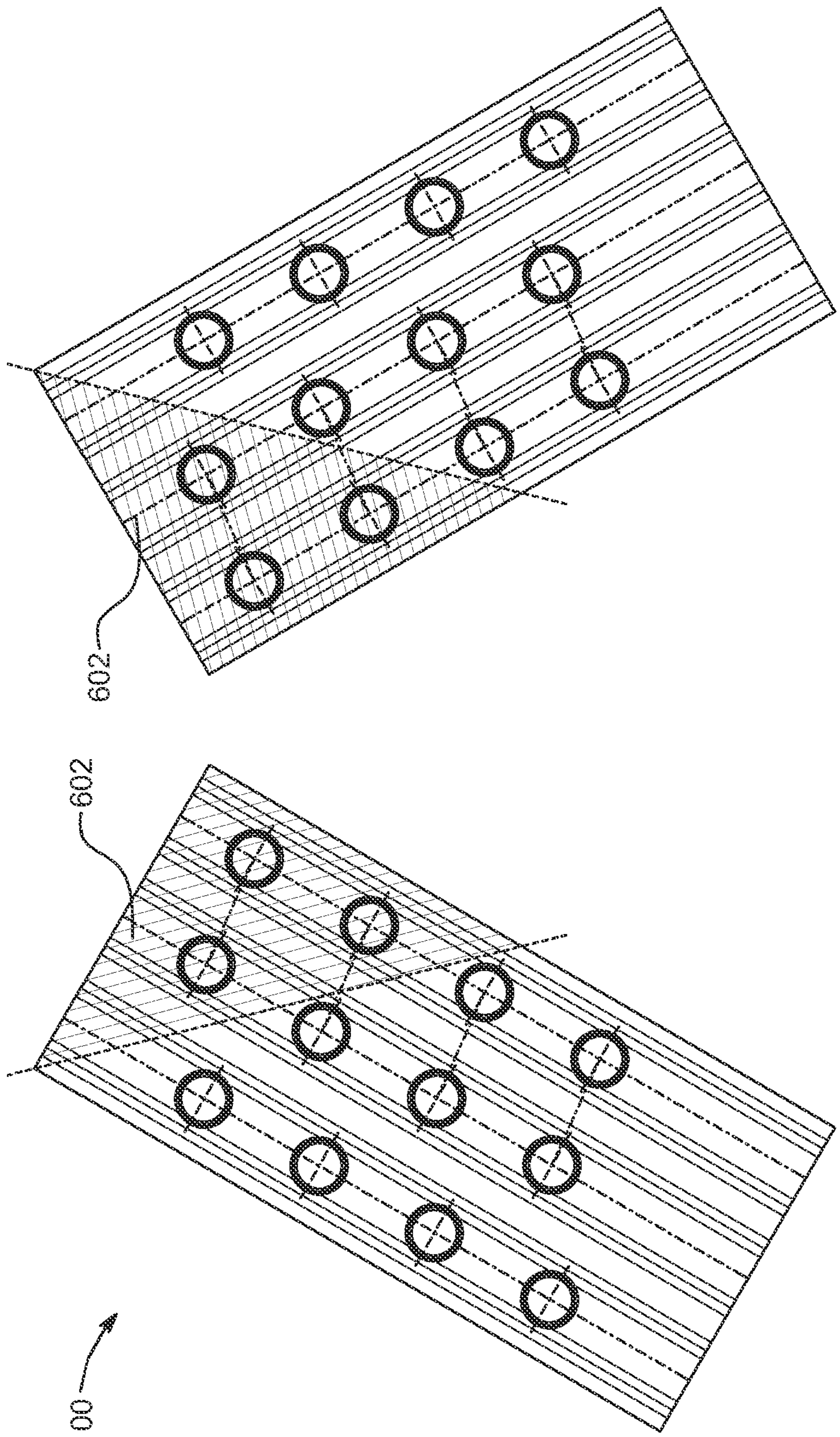

FIG. 6A

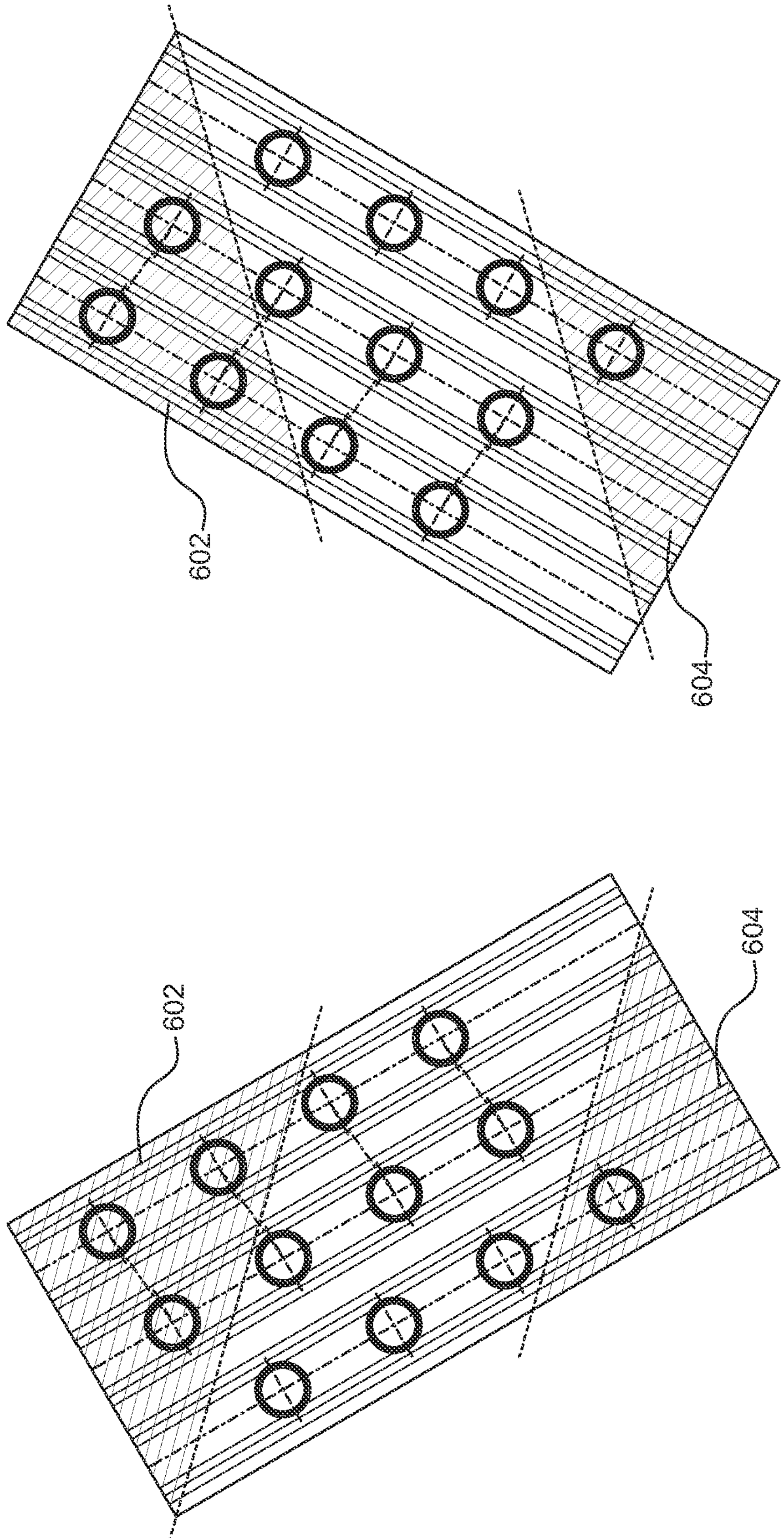

FIG. 6B

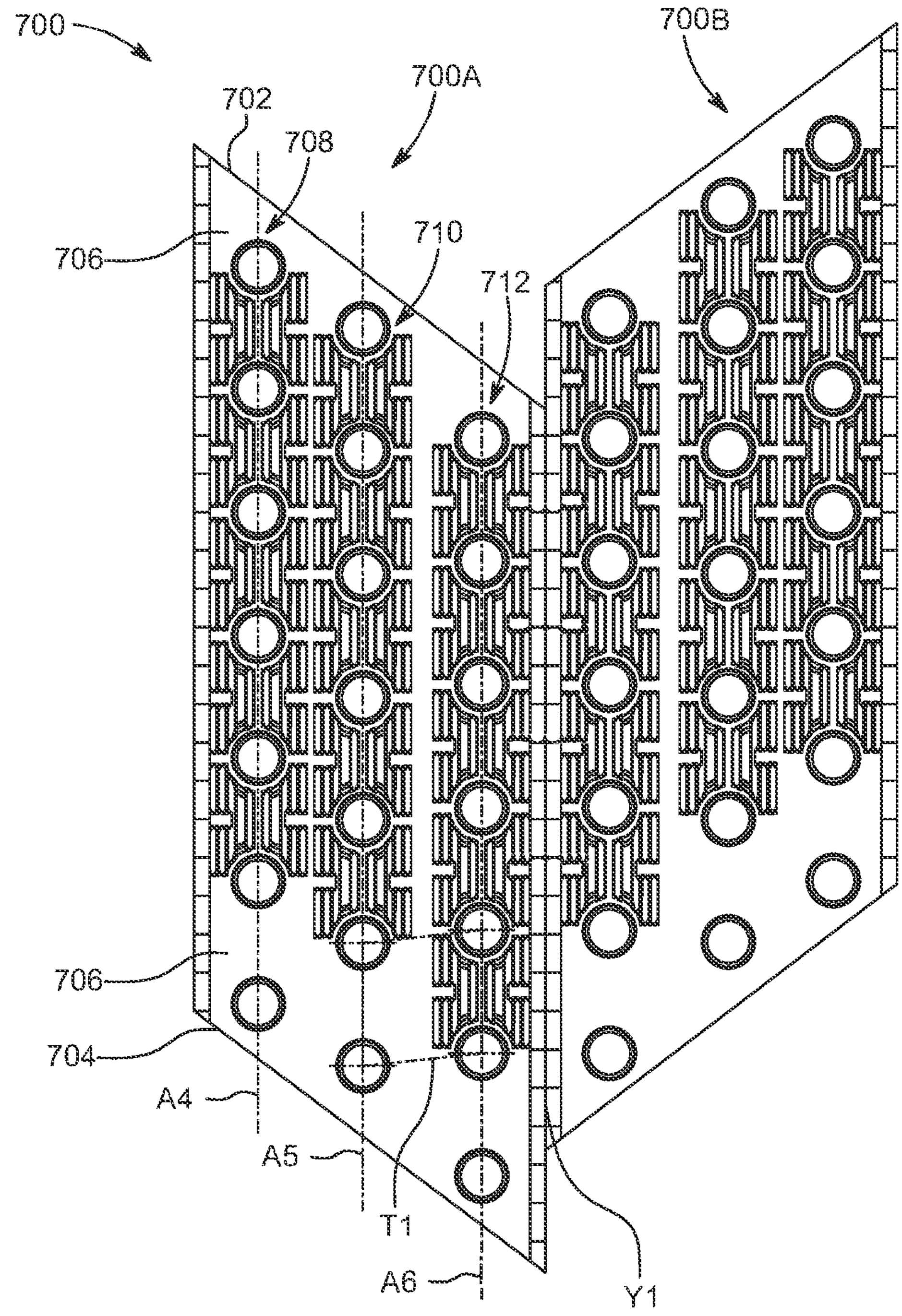

FIG. 7A

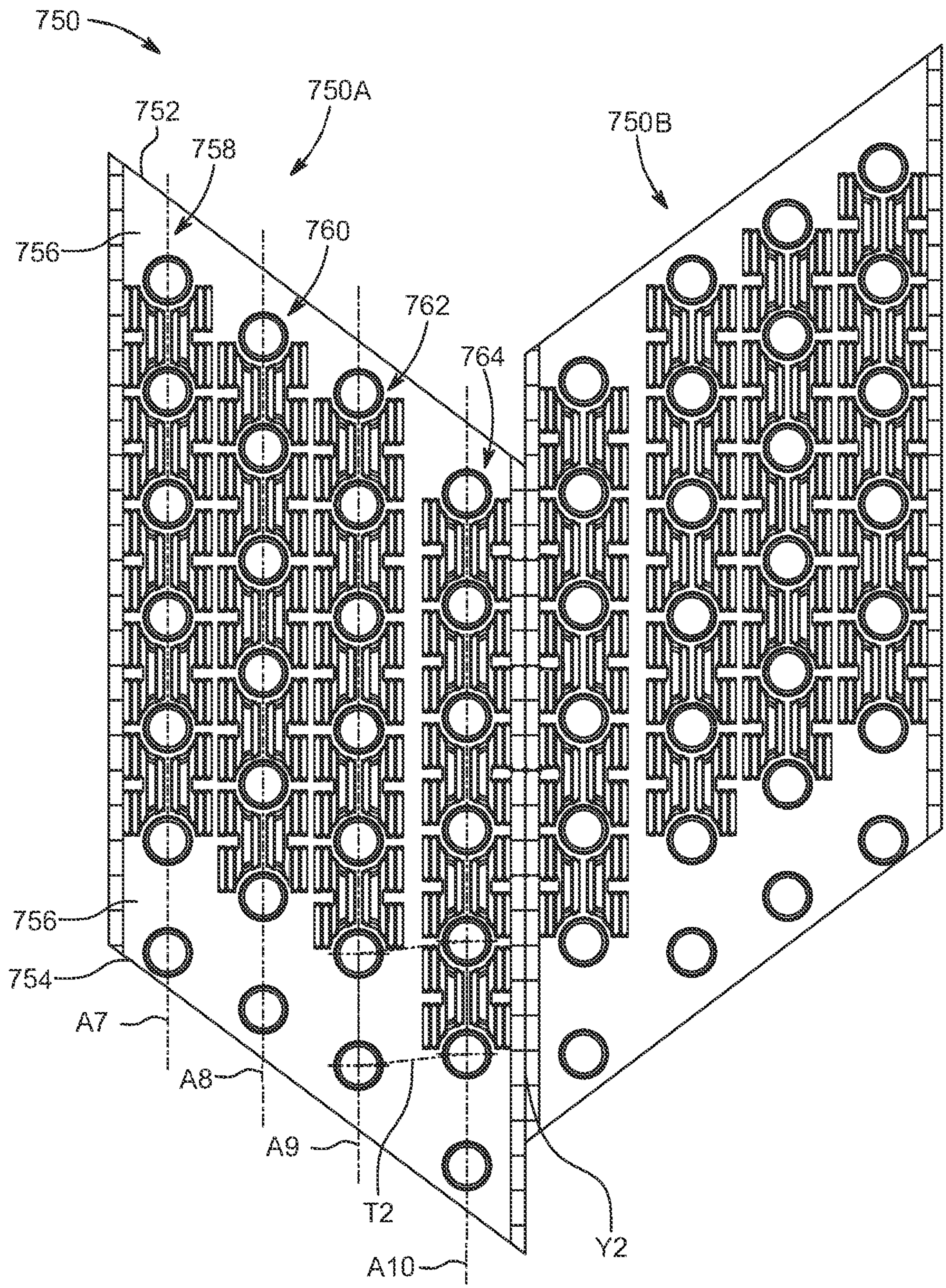

FIG. 7B

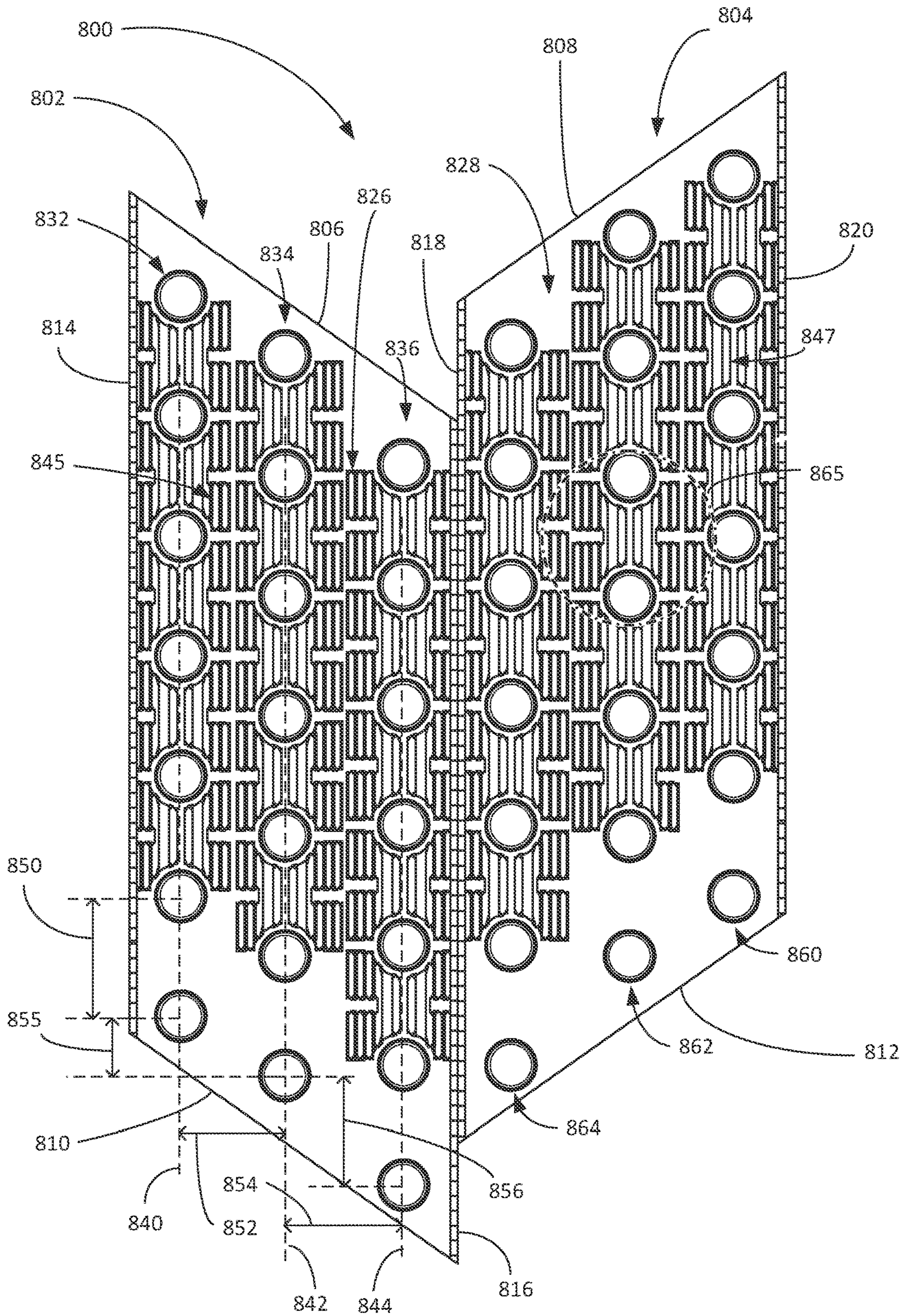
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FIG. 8

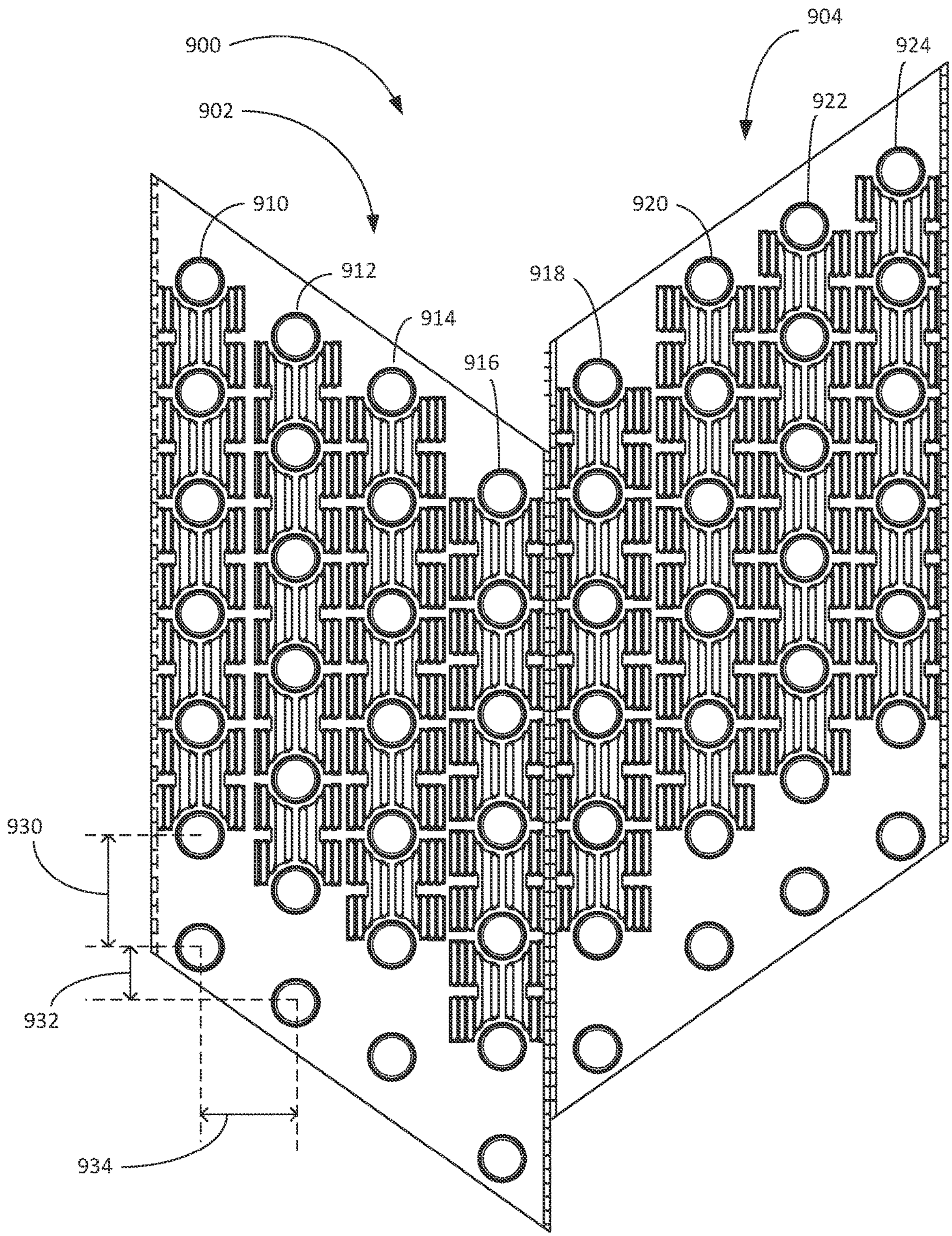


FIG. 9

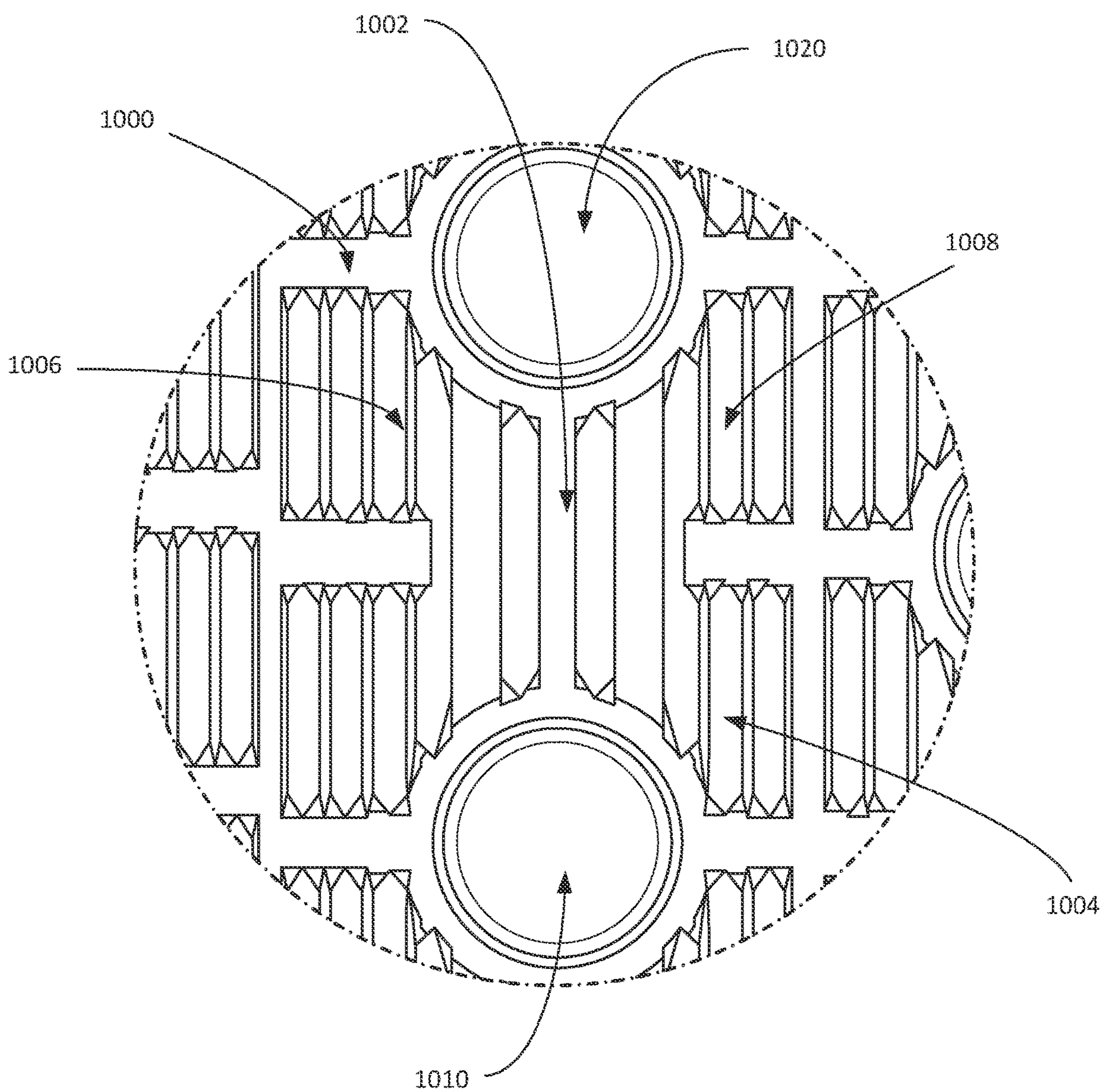
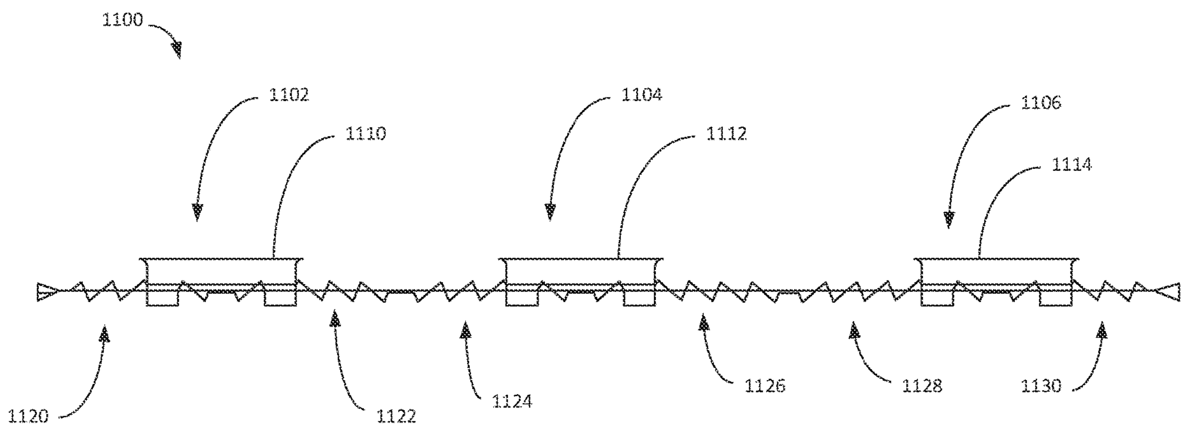


FIG. 10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1102
1110
1104
1112
1106
1114
1120
1122
1124
1126
1128
1130

FIG. 11

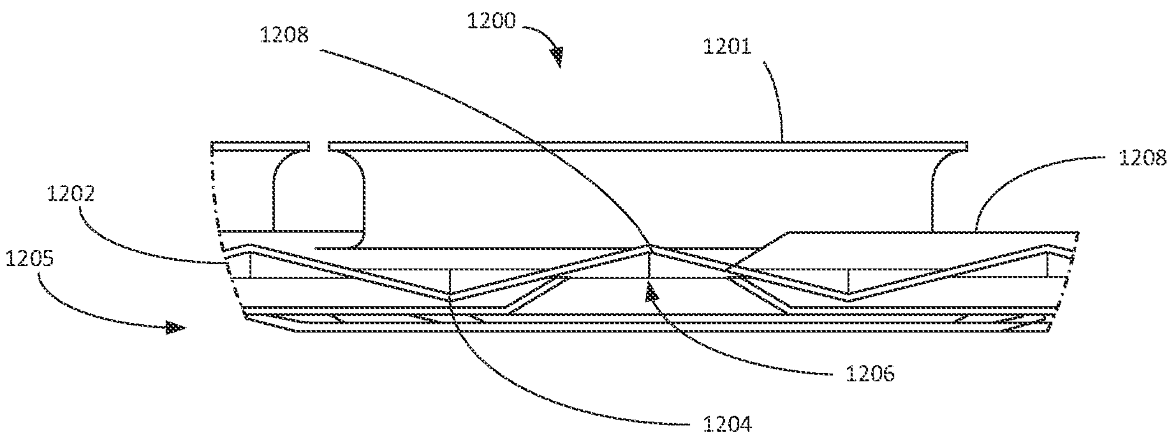

FIG. 12

LOUVERED FIN

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part application of U.S. patent application Ser. No. 17/505,827, filed on Oct. 20, 2021, the entire contents of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates, in general, to a heat exchanger fin and, more specifically relates, to a louvered fin for an evaporator coil.

BACKGROUND

Typically, the condenser coil and the evaporator coil of a heating, ventilation, and air conditioning (HVAC) system are each designed to exchange thermal energy with internal tubing for carrying refrigerant. Each of the condenser coil and the evaporator coil often includes a plurality of fins disposed along a length of the internal tubing, such that adjacent fins are substantially parallel to each other and located apart by a predefined distance. Further, the internal tubing passes through holes defined in the adjacently located fins.

Generally, the condenser coil and the evaporator coil include fins with substantially similar constructional features. Although such similarity in constructional features aid streamlined manufacturing, performance of the fin, such as heat transfer efficiency, may be affected by using similar fins in each of the condenser coil and the evaporator coil. It is required to incorporate condensate management features in the fins implemented in the evaporator coil, while the same may not be a mandate for the condenser coil. As such, design of fins may be made for better performance based on end-use application.

SUMMARY

According to one aspect of the present disclosure, a louvered fin is disclosed. The louvered fin includes a leading edge, a trailing edge opposite to the leading edge, and a surface extending between the leading edge and the trailing edge. The surface defines a plurality of holes, in which a first set of holes are defined along a first axis, a second set of holes are defined along a second axis, and a third set of holes are defined along a third axis. Each of the first axis, the second axis, and the third axis extends substantially parallel to a longitudinal axis of the fin. The second set of holes are offset from the first set of holes along the longitudinal axis of the fin and the third set of holes are offset from the second set of holes along the longitudinal axis of the fin. A first offset distance defined between the second set of holes and the first set of holes is greater than a second offset distance defined between the third set of holes and the second set of holes. The second set of holes and the third set of holes define a substantially obtuse trapezoidal matrix.

In an embodiment, the louvered fin further includes a fourth set of holes defined along a fourth axis extending substantially parallel to the longitudinal axis of the fin.

In an embodiment, the surface extending between the leading edge and the trailing edge is wavy. In an embodiment, each of two opposite wider angles of the obtuse trapezoidal matrix is in a range of about 95 degrees to about 105 degrees.

In an embodiment, a distance between the leading edge and the first axis is in a range of about 0.25 inch to about 0.5 inch. In an embodiment, a distance between the first axis and the second axis is in a range of about 0.50 inch to about 1 inch. In an embodiment, a distance between the third axis and the trailing edge is in a range of about 0.25 inch to about 0.5 inch. In an embodiment, a distance between two adjacent holes of the first set of holes is in a range of about 1 inch to about 0.75 inch.

In an embodiment, a distance between two adjacent holes of the second set of holes is in a range of about 1 inch to about 0.75 inch. In an embodiment, the first offset distance between the first axis and the second axis of holes is in a range of about 0.375 inch to about 0.5 inch. In an embodiment, a diameter of each hole of the first set of holes, the second set of holes, and the third set of holes is in a range of about 6.8 mm to about 4.8 mm.

In an embodiment, the louver fin further includes a plurality of collars, where each collar extends from a periphery of one hole of the plurality of holes. Preferably, each collar extends in a direction perpendicular to the surface of the fin. Each collar includes a narrow portion and an expanded portion, where the expanded portion is distal from the surface of the fin.

According to another aspect of the present disclosure, an evaporator coil is disclosed. The evaporator coil includes a plurality of refrigerant tubes and a plurality of louvered fins. Each louvered fin includes a leading edge, a trailing edge opposite to the leading edge, and a surface extending between the leading edge and the trailing edge. The surface defines a plurality of holes configured to allow the plurality of refrigerant tubes to pass therethrough. A first set of holes of the plurality of holes are defined along a first axis, a second set of holes of the plurality of holes are defined along a second axis, and a third set of holes of the plurality of holes are defined along a third axis. Each of the first axis, the second axis, and the third axis extends substantially parallel to a longitudinal axis of the fin. The second set of holes are offset from the first set of holes along the longitudinal axis of the fin and the third set of holes are offset from the second set of holes along the longitudinal axis of the fin. A first offset distance defined between the second set of holes and the first set of holes is greater than a second offset distance defined between the third set of holes and the second set of holes. The second set of holes and the third set of holes define a substantially obtuse trapezoidal matrix.

In an embodiment, each of the plurality of louvered fins is made from aluminum alloy. In an embodiment, each of the plurality of holes forms an interference fit with an outer surface of a refrigerant tube passing therethrough.

In an embodiment, each of the plurality of louvered fins defines at least one cropped corner. In an embodiment, each of the plurality of louvered fins further includes gagged regions located around peripheries of the hole defined proximal to the at least one cropped corner.

In an embodiment, the evaporator coil further includes a plurality of collars, where each collar extends from a periphery of one hole of the plurality of holes. A length of each collar is in a range of about 1.4 mm to about 1.8 mm.

These and other aspects and features of non-limiting embodiments of the present disclosure will become apparent to those skilled in the art upon review of the following description of specific non-limiting embodiments of the disclosure in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

A better understanding of embodiments of the present disclosure (including alternatives and/or variations thereof) may be obtained with reference to the detailed description of the embodiments along with the following drawings, in which:

FIG. 1A is a front view of a louvered fin, according to an embodiment of the present disclosure;

FIG. 1B is an enlarged view of a portion of the louvered fin of FIG. 1A, according to an embodiment of the present disclosure;

FIG. 1C is a cross-section of a portion of the louvered fin of FIG. 1A, according to an embodiment of the present disclosure;

FIG. 2 is an enlarged portion of the louvered fin of FIG. 1A, according to an embodiment of the present disclosure;

FIG. 3 is a portion of another louvered fin, according to another embodiment of the present disclosure;

FIG. 4 is a side view of the louvered fin of FIG. 3, according to an embodiment of the present disclosure;

FIG. 5 is an enlarged portion of the louvered fin of FIG. 3, according to an embodiment of the present disclosure;

FIG. 6A shows a louvered fin having a cropped corner, according to an embodiment of the present disclosure;

FIG. 6B shows a louvered fin having a pair of cropped corners, according to an embodiment of the present disclosure;

FIG. 7A is a front view of a louvered fin defining three rows of holes, and having cropped corners and gagged regions, according to another embodiment of the present disclosure;

FIG. 7B is a front view of a louvered fin defining four rows of holes, and having cropped corners and gagged regions, according to another embodiment of the present disclosure;

FIG. 8 is a front view of louvered fins each defining three rows of holes, according to another embodiment of the present disclosure;

FIG. 9 is a front view of louvered fins each defining four rows of holes, according to another embodiment of the present disclosure;

FIG. 10 is a focused view the louvers of a louvered fin, according to another embodiment of the present disclosure;

FIG. 11 is a side cross-sectional view of a louvered fin, according to an embodiment of the present disclosure; and

FIG. 12 is a side cross-sectional view of an edge of a louvered fin, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to specific embodiments or features, examples of which are illustrated in the accompanying drawings. Wherever possible, corresponding, or similar reference numbers will be used throughout the drawings to refer to the same or corresponding parts. Moreover, references to various elements described herein, are made collectively or individually when there may be more than one element of the same type. However, such references are merely exemplary in nature. It may be noted that any reference to elements in the singular may also be construed to relate to the plural and vice-versa without limiting the scope of the disclosure to the exact number or type of such elements unless set forth explicitly in the appended claims.

As used herein, the terms “a”, “an” and the like generally carry a meaning of “one or more,” unless stated otherwise. Further, the terms “approximately”, “approximate”, “about”, and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

Aspects of the present disclosure are directed to a louvered fin and an evaporator coil implementing the louvered fin. The louvered fin has a unique design to aid enhanced condensate drainage and enable structural robustness. Additionally, configuration of louvers in the fin form boundary layers of convective heat transfer between air flowing, with an inclined angle of attack, across small diameter tubes extending across the fins.

Referring to FIG. 1A, a front view of a louvered fin **100** is illustrated, according to an embodiment of the present disclosure. The louvered fin **100** (hereinafter referred to as “the fin **100**”) includes a leading edge **102** and a trailing edge **104** opposite to the leading edge **102**. As used herein, the term “leading edge” refers to a feature of the fin **100** that is upstream with respect to a direction of flow of air across the fin **100** and the term “trailing edge” refers to a feature of the fin **100** that is relatively downstream with respect to the direction of flow of air across the fin **100**. These terminologies are well known in the art of fins, for example heat exchanger fins. The fin **100** further includes a surface **106** extending between the leading edge **102** and the trailing edge **104**. Preferably, the surface **106** is wavy in structure. As used herein, the term “wavy” refers to a structure of the fin **100** including adjacent concave and convex portions, or multiple crests and troughs, along a width “W” of the fin **100**. In an embodiment, a length “M” of the fin **100** may be in a range of about 2 inches to 6 inched or 3.8 inches to about 4.5 inches and the width “W” of the fin **100** may be in a range of about 1 inches to 4 inches or 1.9 inches to about 2.5 inches. According to an aspect of the present disclosure, multiple fins, each having a configuration described below, are stacked to constitute a heat exchanging medium of an evaporator coil. The fin **100** may also be suitably implemented in, but not limited to, indoor air handler of residential split systems, both heat pumps and air conditioners.

Further, the surface **106** of the fin **100** defines a plurality of holes including a first set of holes **108**, a second set of holes **110**, and a third set of holes **112**. Each of the plurality of holes is configured to allow a refrigerant tube (not shown) of the evaporator coil to pass therethrough. The phrase “set of holes” may be alternatively referred to and understood as “row of holes”. In an embodiment, a diameter of each hole of the first set of holes **108**, the second set of holes **110**, and the third set of holes **112** is in a range of about 4.8 mm to about 6.8 mm, and preferably 7 mm. According to an aspect of the present disclosure, the fin **100** preferably includes three rows of holes as illustrated in FIG. 1A. The first set of holes **108** are defined along a first axis “A1” located proximal to the leading edge **102** of the fin **100**. In an embodiment, a distance between the leading edge **102** and the first axis “A1” is in a range of about 0.1 inch to about 0.5 inch. In another embodiment, the distance between the leading edge **102** and the first axis “A1” is preferably 0.3625 inch. Further, in an embodiment, a distance between two adjacent holes of the first set of holes **108**, which defines a first pitch “P1”, is in a range of about 0.5 inch to about 1 inch. In some embodiments, preferably, the first pitch “P1” along the first axis “A1” is 0.827 inch.

The second set of holes **110** are defined along a second axis “A2” and are offset from the first set of holes **108** along a longitudinal axis “L” of the fin **100**. The second axis “A2” extends along the longitudinal axis "L" and is located between the first axis "A1" and the trailing edge **104**. For the purpose of brevity, the second axis "A2" is shown coinciding with the longitudinal axis "L". In an embodiment, a distance between the first axis "A1" and the second axis "A2" is in a range of about 0.50 inch to about 1 inch. In another embodiment, distance between the first axis "A1" and the second axis "A2" is preferably 0.725 inch. Further, in an embodiment, a distance between two adjacent holes of the second set of holes **110**, which defines a second pitch "P2", is in a range of about 0.5 inch to about 1 inch. In some embodiments, preferably, the second pitch "P2" along the second axis "A2" is 0.827 inch.

Further, the third set of holes **112** are defined along a third axis "A3" that is located proximal to the trailing edge **104** of the fin **100**. In an embodiment, a distance between the third axis "A3" and the trailing edge **104** is in a range of about 0.1 inch to about 0.5 inch. In another embodiment, the distance between the third axis "A3" and the trailing edge **104** is preferably 0.3625 inch. The third set of holes **112** are offset from the second set of holes **110** along the longitudinal axis "L" of the fin **100**. Further, in an embodiment, a distance between two adjacent holes of the third set of holes **112**, which defines a third pitch "P3", is in a range of about 0.5 inch to about 1 inch. In some embodiments, preferably, the third pitch "P3" along the third axis "A3" is 0.827 inch. Each of the first axis "A1", the second axis "A2", and the third axis "A3" extends substantially parallel to the longitudinal axis "L" of the fin **100**.

The second set of holes **110** are offset at a first offset distance "D1" from the first set of holes **108** and the third set of holes **112** are offset at a second offset distance "D2" from the first set of holes **108**. Preferably, the second offset distance "D2" is less than the first offset distance "D1". In an embodiment, the first offset distance "D1" is in a range of about 0.1 inch to about 0.5 inch. In some embodiment, the first offset distance "D1" preferably is 0.4135 inch. For the purpose of the present disclosure, the offset distances are calculated with respect to centers of the holes as indicated in the FIG. **1**A.

FIG. **1**B is an enlarged view of a portion "B" of FIG. **1**A, according to an embodiment of the present disclosure. As can be seen from FIG. **1**B, the third set of holes **112** are offset from the second set of holes **110** along the longitudinal axis "L". As such, the second set of holes **110** and the third set of holes **112** define a substantially obtuse trapezoidal matrix. An obtuse trapezoid refers to a geometric shape including one acute angle and one obtuse angle defined on a base thereof. For example, centers "C1" of a first hole **110-1**, "C2" of a second hole **110-2**, "C3" of a third hole **112-1**, and "C4" of a fourth hole **112-2** define vertices of the obtuse trapezoid. Line segments "S1" extending between "C1" and "C2", "S2" extending between "C2" and "C3", "S3" extending between "C3" and "C4", and "S4" extending between "C4" and "C1" define sides of the obtuse trapezoid. Line segments "S1" and "S4" define an acute angle "θ1", "S1" and "S2" define an obtuse angle "θ2", and similarly "S3" and "S4" define an obtuse angle "θ4". As such, "θ2" and "θ4" are two opposite wider angles of the obtuse trapezoid. In an embodiment, each of the two opposite wider angles of the obtuse trapezoid is in a range of about 95 degrees to about 105 degrees. Similarly, adjacent set of four holes define another obtuse trapezoid. As such, owing to the offset, the second set of holes **110** and the third set of holes **112** define an array of obtuse trapezoids along the longitudinal axis "L" of the fin **100**, which together constitutes the obtuse trapezoidal matrix.

FIG. **1**C illustrates a cross-section of a portion of the fin **100** considered along a section F-F' in FIG. **1**A. In an embodiment, the fin **100** further includes a plurality of collars, where each collar extends from a periphery of one hole of the plurality of holes. As shown in FIG. **1**C, a collar **114** extends from the periphery of the hole "H" defined in the surface **106** of the fin **100**. Particularly, each collar, such as the collar **114**, extends in a direction perpendicular to the surface **106** of the fin **100**. In an embodiment, the collar **114** includes a narrow portion **116** and an expanded portion **118** extending from the narrow portion **116**. The expanded portion **118** is distal from the surface **106** of the fin **100**. The collar **114** narrows from the periphery of the hole "H" to the narrow portion **116** through a step portion **120**. In order to minimize stress concentration in the collar **114**, the narrow portion **116** and the step portion **120** defines a fillet at a connection portion thereof, having a radius of curvature "R". In an embodiment, a length of each collar is in a range of about 1.4 mm to about 1.8 mm. However, in some embodiments, the length of the collar **114** may be determined based on fins-per-inch factor of the evaporator coil.

According to an aspect, each of the plurality of holes forms an interference fit with an outer surface of the refrigerant tube passing therethrough. For the sake of brevity in illustration and description, the evaporator coil and the refrigerant tubes are not illustrated and particularly described as they are well known in the art. The collar **114**, and other collars formed in the fin **100**, serve to increase mechanical strength of a joinder, such as the interference fit, between the fin **100** and the corresponding refrigerant tube. The collar **114** also serves to increase the heat conductivity between the refrigerant tubes and the fin **100**. In some embodiments, the surface **106** of the fin **100**, the refrigerant tubes, and the collar **106** may each be manufactured from a suitable thermally-conductive material, such as, but not limited to, copper, aluminum, and alloys thereof. In the present disclosure, the fin **100** is made from aluminum alloy.

FIG. **2** illustrates an enlarged portion "G" of FIG. **1**A. Particularly, the enlarged portion "G" includes the hole "H" of the first set of holes **108**. The fin **100** includes a plurality of louvers **202** extending along the longitudinal axis "L" of the fin **100**. In an embodiment, the fin **100** includes a predefined region **204** around the hole "H" configured, for example, to add to the strength of the collar **114** extending from the periphery of the hole "H". As can be seen in FIG. **2**, the predefined region **204** includes a rectangular portion, where a pair of opposite sides of the rectangular portion includes an arcuate portion. Along a horizontal axis "K", the fin **100** includes a sleeve **206** free from the louvers **202**. In other words, the louvers **202** are not formed in the sleeve **206**. Additionally, the predefined region **204** is also free from the louvers **202**. When the refrigerant pipe is inserted through the hole "H", the predefined region **204** and the sleeve **206** may together add strength to the collar **114** extending from the periphery of the hole "H", which otherwise would be weak if the louvers **202** were extending up to the collar **114**. In an example, a width of the sleeve **206** may be about 2 mm. The louvers **202** at extremities, such as a left extreme louver **202-1** and a right extreme louver **202**-N, are wider as compared to intermediate louvers located therebetween. In an example, width of each of louver at the extremities may be about 2.25 mm and width of each intermediate louver may be about 1.5 mm. In some embodiments, the fin **100** may include nine intermediate louvers. FIG. **2** illustrates the louvers **202** associated with the first set of holes **108**. The second set of holes **110** and third set of holes **112** are neighbored by similar set of louvers

202. In some embodiments, number of louvers associated with each row of holes may be equal, for example 11 louvers as illustrated in FIG. 2. In some embodiments, the number of louvers may vary between adjacent rows of holes.

FIG. 3 illustrates a portion of a fin 300, according to another embodiment of the present disclosure. Each of a leading edge 302 and a trailing edge 304 of the fin 300 has a rippled edge, such as a zig-zag pattern (as shown in FIG. 5). A predefined region 306 extends around a circumference of each hole of each row of holes. In an embodiment, louvers 308 are formed to define an X-shape pattern along a surface 310 of the fin 300. In the illustrated embodiment, the fin 300 includes ten louvers 308. In other embodiments, the fin 300 may include any number of louvers 300 based on a width and pitch of the fin 300. Besides being offset, the holes of the second row are defined at a radial distance from the holes of the first row. For example, a distance between center "C5" of a first hole "H1" in the first row of holes and center "C6" of a first hole "H2" in the second row of holes is 21 mm. A circumference of an imaginary circle considered with "C5" as the center, passes through center "C7" of a second hole "H3" in the first set of holes. As such, the center "C7" is also at radial distance from the center "C5", thereby defining the pitch of the fin 300. Similarly, a circumference of another imaginary circle considered with "C6" as the center, passes through center "C8" of a first hole "H4" of the third row of holes and the center "C5" of the first hole "H1" of the first row of holes. As such, the third row of holes are also defined at the same radial distance from the second row of holes. Such design of the fin 300, and that of the fin 100 of FIG. 1A, may eliminate overlap of holes and louvers with adjacent row of holes and louvers, thereby achieving an increase in heat transfer efficiency.

FIG. 4 illustrates a side view of the fin 300. In an embodiment, the fin 300 includes half louvers 402 at the edges of the fin 300, such as the leading edge 302 and the trailing edge 304, and full louvers at remaining portion of the surface 310 of the fin 300. As used herein, the term "full louver" refers to complete length of the louver extending across the surface 310 of the fin 300 and the term "half louver" refers to half length of the louver extending from the surface 310 of the fin 300. FIG. 4 also shows a joint 404 formed between two fins. At the joint 404, the trailing edge of one fin contacts a leading edge of an adjacent fin. Conventional joining processes known to a person skilled in the art to join the rippled edges of the contacting portions of the two fins may be performed to achieve a wider fin array (not shown). In an embodiment, the louvers 308 extend at an angle of about 45 degrees with respect to the surface 310 of the fin 300. In other embodiments, the louvers 308 may be inclined at any preferred acute angle.

FIG. 5 illustrates a portion of the fin 300, according to another embodiment of the present disclosure. The louvers 308 extending to an end of the fin 300 may be subjected to damage or may be directed in random directions during assemblage into the evaporator coil. In order to overcome such damage, in an embodiment, the fin 300 includes gagged regions 502 around the peripheries of the holes defined proximal to the ends of the fin 300. Preferably, the gagged regions 502 are formed around the holes defined proximal to at least one cropped corner 504 in the fin 300. The gagged portions 502 do not include any louvers 308 and hence may add to the strength of the fin 300 at the cropped corner 504.

FIGS. 6A and 6B illustrates fins showing cropped corners. Particularly, FIG. 6A illustrates fins, such as the fin 100, 300, with corners designated for cropping. Hatched portions 602 are indicative of corners that are designated for cropping. The fins are positioned in an inclined manner, for example along arms of alphabet "A". Multiple such fins may be stacked together to constitute a "A-type" evaporator coil. The cropped corners aid in reducing the size of such evaporator coils, in case of space constraints in an end-use application. FIG. 6B illustrates a pair of hatched portions 602, 604 designated for cropping in each fin. The fins in FIG. 6B are oriented in an inclined manner, for example along arms of alphabet "V". Multiple such fins with double cropped corners may be stacked to constitute a smaller evaporator coil based on space available in the end-use application. As such, the fins may be configured to constitute the evaporator coil of desired shape and size. In some embodiments, "N-type", "Z-type", or slab type evaporator coils may be formed in similar manner by positioning the fins, with or without cropped corners, in inclined manner and stacking with refrigerant pipes. However, it may be ensured that cutting line(s) to form the cropped corners does not interfere with the holes defined in the fins and does not lie close to the periphery of the holes. It will be understood to the person skilled in the art that the louvers at the cropped corners need to be handled carefully while stacking multiple such fins to constitute the evaporator coil. In some embodiment, the fins with cropped corners may include gagged regions, such as the regions 502, around the peripheries of the hole defines proximal to the cropped corners. In some embodiments, the fin may include two or more cropped corners.

FIG. 7A illustrates a front view of a louvered fin 700, according to another embodiment of the present disclosure. Preferably, two fins 700A and 700B may be die stamped on an aluminum sheet to achieve the design illustrated in FIG. 7A. Further, the fin 700A may be separated from the fin 700B, by known means, along a line of contact "Y1". Upon separation, multiple such fins may be disposed adjacent to each other and may be stacked together with refrigerant tubes. Each of the fins 700A, 700B includes three rows of holes defined between cropped corners 702, 704, and gagged regions 706 around holes defined proximal to the cropped corners 702, 704. For the purpose of brevity, configuration of the fin 700A is described herein. A first row of holes 708 is defined along a first axis "A4", a second row of holes 710 is defined along a second axis "A5", and a third row of holes 712 is defined along a third axis "A6". Each axis extends substantially parallel to a longitudinal axis of the fin 700A. The second row of holes 710 are offset from the first row of holes 708 in a direction along a length of the fin 700A, and the third row of holes 712 are offset from the second row of holes 710 in the direction along the length of the fin 700A. The second row of holes 710 and the third row of holes define a substantially obtuse trapezoidal matrix. For the purpose of illustration, one obtuse trapezoid "T1" is indicated in fin 700A. Interior angles each obtuse trapezoid formed in the fin 700A may preferably be equal to those described with respect to FIG. 1B.

FIG. 7B illustrates a front view of a louvered fin 750, according to another embodiment of the present disclosure. Preferably, two fins 750A and 750B may be die stamped on an aluminum sheet to achieve the design illustrated in FIG. 7B. Further, the fin 750A may be separated from the fin 750B, by known means, along a line of contact "Y2". Upon separation, multiple such fins may be disposed adjacent to each other and may be stacked together with refrigerant tubes. Each of the fins 750A, 750B includes four rows of holes defined between cropped corners 752, 754, and gagged regions 756 around holes defined proximal to the cropped corners 752, 754. For the purpose of brevity, configuration of the fin **750**A is described herein. The fin **750**A includes a first row of holes **758** defined along a first axis "A7", a second row of holes **760** defined along a second axis "A8" and offset from the first row of holes **758**, a third row of holes **762** defined along a third axis "A9" and offset from the second row of holes **760**, and a fourth row of holes **764** defined along a fourth axis "A10" and offset from the third row of holes **762**. Each axis extends substantially parallel to a longitudinal axis of the fin **750**A. Geometrically, the third row of holes **760** and the fourth row of holes **762** define a substantially obtuse trapezoidal matrix. For the purpose of illustration, one obtuse trapezoid "T2" is indicated in fin **750**A. Interior angles each obtuse trapezoid formed in the fin **750**A may preferably be equal to those described with respect to FIG. **1**B.

FIG. **8** illustrates a front view of fins **800**, according to another embodiment of the present disclosure. Specifically, fins **800** may include fin **802** and fin **804**, which may be positioned adjacent to one another as shown in in FIG. **8**. For example, fins **802** and **804** may be standalone or fins **802** and **804** may be joined or otherwise connected (e.g., via welding, bolted connection, threaded connection, or the like or may be made from the same integral piece). It is understood that fins **802** and **804** may be positioned next to fins that are identical or similar to fins **802** and **804**. For example, multiple such fins may be disposed adjacent to each other and may be stacked together with refrigerant tubes.

Fins **802** and **804** may be die stamped on a metallic (e.g., aluminum) sheet to achieve the design illustrated in FIG. **8**. However, it is understood that different manufacturing techniques may be used to created fins **802** and **804** (e.g., casting). It is further understood that fins **802** and **804** may be made of a different material (e.g., copper, steel, alloy, etc.). In one example, fins **802** and **804** may be die stamped together and/or may be simultaneously or subsequently separated.

Fin **802** and fin **804** may have leading edge **806** and leading edge **808**, respectively. Leading edge **806** and leading edge **808** may be linear and may be angled. Fin **802** and fin **804** may similarly have trailing edge **810** and trailing edge **812**, respectively. Trailing edge **806** and trailing edge **808** may be linear and may be parallel to leading edge **806** and leading edge **808**, respectively. It is understood that leading edge **806** and leading edge **808** may not be parallel. Fin **802** may further include left edge **814** and right edge **816** that are linear and parallel. Fin **804** may further include left edge **818** and right edge **820** that are linear and parallel. Fin **802** and fin **804** may each be parallelograms and/or may be mirror images or transpositions of one another.

As shown in FIG. **8**, fin **802** and fin **804** may each include three sets of holes arranged in three columns. Each set of holes may include several holes equally spaced along an axis. Each axis of each set of holes may be parallel with each other axis of each set of holes. Each axis of each set of holes may also be parallel to left edge **814**, right edge **816**, left edge **818** and right edge **820**.

Each hole of the holes in fin **802** and **804** may be the same or similar to first hole **110-1** of FIG. **1**B and/or hole **1110**, hole **1112** and/or hole **1114** of FIG. **11**. Each hole of the set of holes in fins **802** and **804** may be designed to receive one or more heat exchange tube (e.g. a refrigerant tube) that may be designed to receive and circulate a fluid (e.g., a refrigerant) and may exchange thermal energy with the heat exchange tube and/or fluid.

Surface **826** may be defined between leading edge **806**, trailing edge **810**, left edge **814** and right edge **816**. Surface **828** may be defined between leading edge **808**, trailing edge **812**, left edge **818**, and right edge **820**. Surface **826** may include louvers **845** that my extend entirely between left edge **814** and right edge **816** and may extend partially between leading edge **806** and trailing edge **810**. Surface **828** may include louvers **847** that may extend entirely between left edge **818** and right edge **820** and may extend partially between leading edge **808** and trailing edge **812**. Louvers **845** and/or **847** may be the same or similar to louver **308** of FIG. **3**. Louvers in region **865** are illustrated in greater detail in FIG. **10**.

Fin **802** may include set of holes **832** which may include several holes arranged along axis **840**, set of holes **834**, which may be arranged along axis **842**, and set of holes **836**, which may be arranged along axis **844**. Each of set of holes **832**, set of holes **834**, and set of holes **836** may include the same number of holes and may include holes that are equally distributed along axis **840**, **842**, and **844**, respectively and separated along each respective axis by the same distance. For each hole of the set of holes **832**, set of holes **834**, and set of holes **836** may be spaced apart along each respective axis **840**, axis **842**, and axis **846** by an amount **850**. In one example, amount **850** may be between 0.75 and 1 inch.

Axis **840** of set of holes **832** may be horizontally offset from axis **842** of set of holes **834** by value **852**. In one example, value **852** may be between 0.5 and 0.75 inch. Similarly, axis **842** of set of holes **834** may be offset from axis **844** of set of holes **836** by valve **854**. In one example, value **854** may be between 0.75 and 1 inch. It is understood that value **854** may be the same or substantially the same (e.g., within a manufacturing tolerance) of value **850**. With value **854** and value **850** being the same or substantially the same, similar tooling and parts may be used between the respective holes of fin **802**.

Holes of set of holes **832** may be vertically offset from holes of set of holes **834** by value **855**. Holes of set of holes **834** may be vertically offset from holes of set holes **836** by value **856**. Value **856** may be larger than value **855**. It is understood that value **856** may be calculated based on value **850**, value **852**, and/or value **855**. It is further understood that value **854** may be larger than value **852** and that more louver peaks and/or valleys may exist between set of holes **834** and set of holes **836** than between set of holes **832** and set of holes **834**. In one example, two consecutive holes of set of holes **834** may together with two counterpart holes of set of holes **834** may define a substantially obtuse trapezoidal matrix. Similarly, two consecutive holes of set of holes **862** may together with two counterpart holes of set of holes **864** may define a substantially obtuse trapezoidal matrix of fin **804**.

Fin **804** may be designed such that it mirrors or transposes the design of fin **802**. For example, fin **804** may similarly include set of holes **860**, which may be similar to set of holes **832**, set of holes **862**, which may be similar to set of holes **834**, and set of holes **864**, which may be similar to set of holes **836**. The holes of set of holes **860**, **862** and **864** may each be equally distanced along respective parallel axes spaced apart by value **850**. It is understood that set of holes **864** may be positioned between set of holes **862** and set of holes **836**. It is further understood that at least some holes of set of holes **864** and set of holes **836** may be horizontally aligned.

Set of holes **860** may be offset horizontally from set of holes **862** by value **854**. Set of holes **862** may be offset horizontally from set of holes **864** by amount **854**. In one example, value **854** may be between 0.5 and 1.0 inch. Holes of set of holes **860** may be vertically offset from holes of set of holes **862** by value **855**. In one example, value **855** may be between 0.25 and 0.5 inch. Holes of set of holes **862** may be vertically offset from holes of set holes **864** by value **856**. Value **856** may be larger than value **855**.

As shown in FIG. **8**, louvers **845** and louvers **847** may only extend across a portion of fin **802** and fin **804**, respectively. For example, louvers **845** may extend between left edge **814** and right edge **816** of fin **802** and may extend between left edge **818** and right edge **820** of fin **804**. However, louvers **845** of fin **802** may not extend all the way to leading edge **806** and/or trailing edge **810** and louvers **847** of fin **804** may not extend all the way to leading edge **801** and trailing edge **812**.

For example, set of holes **832** may include a hole nearest trailing edge **810** and a hole second nearest trailing edge **810**, set of holes **834** may include a hole nearest trailing edge **810** and a hole second nearest trailing edge **810**, and set of holes **836** may include a hole nearest trailing edge **810** and a hole second nearest trailing edge **810**. As shown in FIG. **8**, louvers **845** may terminate at the holes second nearest trailing edge **810** such that surface **825** is without louvers at the holes of set of holes **832**, **834**, **836** nearest trailing edge **810**. Surface **825** without louvers may more easily guide condensation to and past trailing edge **810** and ultimately off of fin **802**. It is understood that louvers **845** may alternatively terminate immediately prior to or immediately after the holes of second nearest trailing edge **810**. It is further understood that louvers **847** may similarly terminate prior to leading edge **808** and trailing edge **812**.

Each of left edge **814**, right edge **816**, left edge **818**, and right edge **820** may include a rippled edge, such as a zig-zag pattern (as shown in FIG. **5** and/or FIG. **12**). For example, the rippled edge may include alternating peaks and valleys. It is understood that the peaks and valleys along each right edge and trailing edge may be oriented perpendicular to the peaks and valleys of louvers **845** and louvers **847**.

FIG. **9** illustrates a front view of fins **900**, according to another embodiment of the present disclosure. Specifically, fins **900** may include fin **902** and fin **904**, which may be positioned adjacent to one another as shown in in FIG. **9**. Fins **902** and **904** may be the same as or substantially the same as fins **802** and **804**, except that each of fins **902** and **904** may include four sets of holes arranged in four columns.

Fin **902** may include set of holes **910**, set of holes **912**, set of holes **914**, and set of holes **916**. Set of holes **912**, set of holes **914**, and set of holes **916** may be the same as or substantially the same as set of holes **832**, set of holes **834**, and set of holes **836** of FIG. **8** and may include the same spacing and arrangement as set of holes **832**, set of holes **834**, and set of holes **836**. Fin **902** may further include set of holes **910** which may be arranged along an axis parallel to that of set of holes **912**.

Set of holes **910** may include several holes that are equally spaced apart. For example, each hole may be spaced apart by amount **930**, which may be the same as amount **850** of FIG. **8**. Holes of set of holes **910** may be vertically offset from holes of set of holes **912** by value **932**, which may be the same as value **855** of FIG. **8**. Holes of set of holes **910** may be horizontally spaced apart from holes of set of holes **912** by value **934**, which may be the same as value **852** of FIG. **8**

Fin **904** may be designed such that it mirrors or transposes the design of fin **902**. For example, fin **904** may similarly include set of holes **924**, which may be similar to set of holes **910**, set of holes **922**, which may be similar to set of holes **912**, set of holes **920**, which may be similar to set of holes **914**, and set of holes **918**, which may be similar to set of holes **916**. The holes of set of holes **924**, **922**, **920**, and **918** may each be distanced along respective parallel axes spaced apart by value **930**. It is understood that set of holes **918** may be positioned between set of holes **918** and set of holes **916**. It is further understood that at least some holes of set of holes **916** and set of holes **918** may be horizontally aligned.

FIG. **10** illustrates region **1000**, which may be the same as or similar to **865** of FIG. **8**, according to another embodiment of the present disclosure. Region **1000** may extend around a circumference of a hole in the set of holes. For example, region **1000** may surround a hole in set of holes **836** and/or set of holes **862** of FIG. **8** and/or set of holes **916** and/or set of holes **920** of FIG. **9**. For example, louvers **1004** may be formed between hole **1010** and hole **1020** to define an X-shape pattern of louvers along a surface **1000** of the fin.

In one example, the fin may include eleven columns of louvers **1004**. For example, the X-shape may include left side **1006**, which may include six columns of louvers **1004** and right side **1008** which may include five columns of louvers. In the embodiment illustrated in FIG. **10**, holes **1010** and **1020** may be offset a greater distance from a set of holes on the left-hand side than a set of holes on the right hand side. As a result, more columns of louvers **1004** may be positioned on left side **1006** of louvers **1004** than right side **1008**.

It is understood that louvers **1004** may be a different shape instead of X-shaped. For example, louvers **1004** may form continuous columns, may extend horizontally instead of vertically, or may be concentric circles. It is further understood that right side **1008** of louvers **1004** may include the same number or a greater number of columns than left side **1006**.

FIG. **11** illustrates a cross-sectional view of a fin and three holes, according to another embodiment of the present disclosure. Specifically, fin **1100** includes hole **1110** of set of holes **1102**, hole **1112** of set of holes **1104**, and hole **1114** of set of holes **1106**. Fin **1100** may be the same or similar to fin **802** of FIG. **8**, for example. Set of holes **1102** may be the same or similar to set of holes **832**, set of holes **1104** may be the same or similar to set of holes **834**, and set of holes **1106** may be the same or similar to set of holes **836** of FIG. **8**.

As shown in FIG. **11**, hole **1112** may be positioned closer to hole **1110** than to hole **1114**. Hole **1110** may be surrounded by louvers **1120** and **1122**, which may include alternating peaks and valleys. Similarly hole **1112** may be surrounded by louvers **1124** and **1126**, which may include alternating peaks and valleys. While louver **1124** may be similar to louver **1120**, unlike louver **1122** on the right side of hole **1110**, which includes three peaks and three valleys, louver **1126** on the right side if hole **1112** may include four peaks and four valleys. Hole **1114** may be surrounded by louvers **1128**, which may include four peaks and four valleys, and louver **1130**, which may include three alternating peaks and valleys.

As shown in FIG. **11**, the louvers may be slanted or biased towards each respective hole. For example, louvers **1126** may be slanted towards hole **1112** and louvers **1128** may be slanted towards hole **1114**. It is understood that any other peak and/or valley design may be used, including different pitch, angles, depth, height, number of columns, width, and the like.

FIG. **12** illustrates a cross-sectional view of a rippled edge of a fin (e.g., left edge or right edge), according to another embodiment of the present disclosure. Specifically, edge **1205** may be the left edge or right edge of a fin extending between a trailing edge and a leading edge. Edge **1205** may include ripple **1202** which may include alternating peaks **1208** and valleys **1204**. Alternating peaks **1208** and valleys

1204 may be similar to louvers (e.g., louvers **1128** of FIG. **11**) in that each may increase the surface area for a given portion of the fin, which may facilitate thermal energy transfer with the exterior environment in which the fin is situated (e.g., air).

Hole **1200** may be positioned adjacent to and interior to ripple **1202**. Hole **1201** may extend from and through surface **1206**. Ripple **1202** may also extend from surface **1206**. As shown in FIG. **12**, louvers **1208** may also extend from surface **1206** and may be positioned between ripple **1202** and hole **1201**. In one example, louver **1208** may be the same as or similar to louvers **1004** of FIG. **10**.

To this end, the present disclosure provides a unique design for louvered fins. The louvered fin is particularly designed for condensate draining, utilizing larger louver angle, but implementing plurality of louvers in contrast to conventional lanced fin. With such configuration of the louvered fin, it is possible to maintain a high convective heat transfer coefficient. Besides contributing for good condensate drainage, the louvered fin of the present disclosure also restarts boundary layers of convective heat transfer with small diameter refrigerant tubes, for example 7 mm tubes, at an inclined angle of attack for the airflow across the louvered fins. Additionally, the geometrical characteristics of the louvered fin, such as the obtuse trapezoidal matrix of the holes, may aid easy draining of condensate.

While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed features without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

What is claimed is:

1. A heat exchange fin comprising:
a leading edge;
a trailing edge opposite to the leading edge;
a surface extending between the leading edge and the trailing edge, the surface defining a plurality of holes, the surface comprising:
a first edge and a second edge opposite the first edge, the first edge and the second edge different from the leading edge and the trailing edge, wherein each of the first edge and the second edge comprises alternating peaks and valleys;
a first set of holes of the plurality of holes defined along a first axis;
a second set of holes of the plurality of holes defined along a second axis, parallel to the first axis;
a first number of fins disposed between the first set of holes and the second set of holes;
a third set of holes of the plurality of holes defined along a third axis, parallel to the second axis, the second set of holes positioned between the third set of holes and the first set of holes; and
a second number of fins disposed between the second set of holes and the third set of holes, wherein the second number is greater than the first number;
wherein each of the holes of the first set of holes, the second set of holes, and the third set of holes are spaced apart along a first direction parallel to the first axis by a first amount;
wherein the first set of holes and the second set of holes are offset along a second direction perpendicular to the first direction by a second amount; and
wherein the third set of holes is offset from the second set of holes along the second direction by a third amount different than the second amount.

2. The heat exchange fin of claim **1**, wherein the second set of holes and the third set of holes define a substantially obtuse trapezoidal matrix.

3. The heat exchange fin of claim **1**, wherein the surface further comprises a fourth set of holes of the plurality of holes defined along a fourth axis, parallel to the first axis.

4. The heat exchange fin of claim **3**, wherein the fourth set of holes of the plurality of holes is positioned such that the first set of holes is positioned between the fourth set of holes and the second set of holes.

5. The heat exchange fin of claim **4**, wherein the fourth set of holes is offset from the first set of holes along the second direction by the second amount.

6. The heat exchange fin of claim **1**, wherein the third set of holes is offset from the second set of holes along the first direction by a fourth amount different from the first amount.

7. The heat exchange fin of claim **1**, wherein the first set of holes comprises a first hole near the trailing edge and a second hole between the first hole and the trailing edge, the second set of holes comprises a third hole near the trailing edge and a fourth hole between the third hole and the trailing edge, and the third set of holes comprises a fifth hole near the trailing edge and a sixth hole between the first hole and the trailing edge, and wherein the fins do not extend between the first hole and the trailing edge, the third hole and the trailing edge, and the fifth hole and the trailing edge.

8. The heat exchange fin of claim **1**, wherein each hole of the plurality of holes is configured to receive one or more refrigerant tubes and the surface is configured to exchange thermal energy with the one or more refrigerant tubes.

9. The heat exchange fin of claim **1**, wherein a first distance between a center of a first hole of the first set of holes and a center of a second hole of the second set of holes is equal to a second distance between the center of the second hole and a center of a third hole of the third set of holes.

10. The heat exchange fin of claim **1**, wherein a circumference of an imaginary circle with a center at a center of a first hole of the second set of holes passes through a center of a second hole of the first set of holes and passes through a center of a third hole in the third set of holes.

11. The heat exchange fin of claim **1**, wherein the first number of fins and the second number of fins are louvered fins.

12. The heat exchange fin of claim **1**, wherein the surface comprises fins extending across a majority of the surface.

13. A heat exchanger comprising:
one or more refrigerant tubes configured to receive and guide a fluid;
a heat exchange fin configured to exchange thermal energy with the one or more refrigerant tubes, the heat exchange fin comprising:
a leading edge;
a trailing edge opposite to the leading edge;
a surface extending between the leading edge and the trailing edge, the surface defining a plurality of holes, each hole of the plurality of holes sized and configured to receive the one or more refrigerant tubes, the surface comprising:
a first set of holes of the plurality of holes defined along a first axis;
a second set of holes of the plurality of holes defined along a second axis, parallel to the first axis;

a third set of holes of the plurality of holes defined along a third axis, parallel to the second axis, the second set of holes positioned between the third set of holes and the first set of holes;

wherein the first set of holes comprises a first hole near the trailing edge and a second hole between the first hole and the trailing edge, the second set of holes comprises a third hole near the trailing edge and a fourth hole between the third hole and the trailing edge, and the third set of holes comprises a fifth hole near the trailing edge and a sixth hole between the first hole and the trailing edge, and wherein the fins do not extend between the first hole and the trailing edge, the third hole and the trailing edge, and the fifth hole and the trailing edge;

wherein each of the holes of the first set of holes, the second set of holes, and the third set of holes are spaced apart along a first direction parallel to the first axis by a first amount;

wherein the first set of holes and the second set of holes are offset along a second direction perpendicular to the first direction by a second amount; and

wherein the third set of holes is offset from the second set of holes along the second direction by a third amount different than the second amount, and the second set of holes is offset from the third set of holes along the first direction by a fourth amount different from the first amount.

14. The heat exchanger of claim **13**, further comprising:

a second heat exchange fin configured to exchange thermal energy with the one or more refrigerant tubes, the second heat change fin comprising:

a second leading edge;

a second trailing edge opposite to the second leading edge;

a second surface extending between the second leading edge and the second trailing edge, the second surface defining a second plurality of holes, each hole of the second plurality of holes sized and configured to receive the one or more refrigerant tubes, the second surface comprising:

a primary set of holes of the plurality of holes defined along a primary axis;

a secondary set of holes of the plurality of holes defined along a secondary axis, parallel to the primary axis;

a tertiary set of holes of the plurality of holes defined along a tertiary axis, parallel to the secondary axis, the secondary set of holes positioned between the tertiary set of holes and the primary set of holes;

wherein each of the holes of the primary set of holes, the secondary set of holes, and the tertiary set of holes are spaced apart along the first direction by the first amount;

wherein the primary set of holes and the secondary set of holes are offset along the second direction perpendicular to the first direction by the second amount; and

wherein the tertiary set of holes is offset from the secondary set of holes along the second direction by the third amount different than the second amount.

15. The heat exchanger of claim **14**, wherein the second heat exchange fin is positioned adjacent to the first heat exchange fin.

16. The heat exchanger of claim **15**, wherein the primary axis is parallel to the first axis, and wherein the tertiary set of holes is positioned immediately between the secondary set of holes and the third set of holes.

17. The heat exchanger of claim **13**, wherein the second set of holes and the third set of holes define a substantially obtuse trapezoidal matrix.

18. The heat exchanger of claim **13**, wherein the surface further comprises a fourth set of holes of the plurality of holes defined along a fourth axis, parallel to the first axis, and the fourth set of holes of the plurality of holes is positioned such that the first set of holes is positioned between the fourth set of holes and the second set of holes.

19. The heat exchanger of claim **18**, wherein the fourth set of holes is offset from the first set of holes along the second direction by the second amount.

20. A heat exchange fin comprising:

a leading edge;

a trailing edge opposite to the leading edge;

a surface extending between the leading edge and the trailing edge, the surface defining a plurality of holes, the surface comprising:

a first set of holes of the plurality of holes defined along a first axis;

a second set of holes of the plurality of holes defined along a second axis, parallel to the first axis;

a third set of holes of the plurality of holes defined along a third axis, parallel to the second axis, the second set of holes positioned between the third set of holes and the first set of holes,

wherein each of the holes of the first set of holes, the second set of holes, and the third set of holes are spaced apart along a first direction parallel to the first axis by a first amount; and

wherein the first set of holes and the second set of holes are offset along a second direction perpendicular to the first direction by a second amount wherein the third set of holes is offset from the second set of holes along the second direction by a third amount different than the second amount, the third amount is greater than the second amount.

21. The heat exchange fin of claim **20**, wherein the first set of holes is offset from the second set of holes along the first direction by a fourth amount,

wherein the second set of holes is offset from the third set of holes along the first direction by a fifth amount, wherein the fifth amount is greater than the fourth amount.

22. A heat exchange fin comprising:

a leading edge;

a trailing edge opposite to the leading edge;

a surface extending between the leading edge and the trailing edge, the surface defining a plurality of holes, the surface comprising:

a first set of holes of the plurality of holes defined along a first axis;

a second set of holes of the plurality of holes defined along a second axis, parallel to the first axis;

a first number of fins disposed between the first set of holes and the second set of holes;

a third set of holes of the plurality of holes defined along a third axis, parallel to the second axis, the second set of holes positioned between the third set of holes and the first set of holes; and

a second number of fins disposed between the second set of holes and the third set of holes, wherein the second number is greater than the first number;

wherein a first distance between a center of a first hole of the first set of holes and a center of a second hole of the second set of holes is equal to a second distance between the center of the second hole and a center of a third hole of the third set of holes;

wherein each of the holes of the first set of holes, the second set of holes, and the third set of holes are spaced apart along a first direction parallel to the first axis by a first amount;

wherein the first set of holes and the second set of holes are offset along a second direction perpendicular to the first direction by a second amount; and

wherein the third set of holes is offset from the second set of holes along the second direction by a third amount different than the second amount.

23. A heat exchange fin comprising:

a leading edge;

a trailing edge opposite to the leading edge;

a surface extending between the leading edge and the trailing edge, the surface defining a plurality of holes, the surface comprising:

a first set of holes of the plurality of holes defined along a first axis;

a second set of holes of the plurality of holes defined along a second axis, parallel to the first axis;

a first number of fins disposed between the first set of holes and the second set of holes;

a third set of holes of the plurality of holes defined along a third axis, parallel to the second axis, the second set of holes positioned between the third set of holes and the first set of holes; and

a second number of fins disposed between the second set of holes and the third set of holes, wherein the second number is greater than the first number;

wherein a circumference of an imaginary circle with a center at a center of a first hole of the second set of holes passes through a center of a second hole of the first set of holes and passes through a center of a third hole in the third set of holes;

wherein each of the holes of the first set of holes, the second set of holes, and the third set of holes are spaced apart along a first direction parallel to the first axis by a first amount;

wherein the first set of holes and the second set of holes are offset along a second direction perpendicular to the first direction by a second amount; and

wherein the third set of holes is offset from the second set of holes along the second direction by a third amount different than the second amount.

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